

A two-dimensional model of hydraulic performance of stormwater infiltration systems

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Abstract:

Stormwater infiltration systems are a popular method for urban stormwater control. They are often designed using an assumption of one-dimensional saturated outflow, although this is not very accurate for many typical designs where two-dimensional (2D) flows into unsaturated soils occur. Available 2D variably saturated flow models are not commonly used for design because of their complexity and difficulties with the required boundary conditions. A purpose-built stormwater infiltration system model was thus developed for the simulation of 2D flow from a porous storage. The model combines a soil moisture-based model for unsaturated soils with a ponded storage model and uses a wetting front-tracking approach for saturated flows. The model represents the main physical processes while minimizing input data requirements. The model was calibrated and validated using data from laboratory 2D stormwater infiltration trench experiments. Calibrations were undertaken using five different combinations of calibration data to examine calibration data requirements. It was found that storage water levels could be satisfactorily predicted using parameters calibrated with either data from laboratory soils tests or observed water level data, whereas the prediction of soil moistures was improved through the addition of observed soil moisture data to the calibration data set. Copyright © 2012 John Wiley & Sons, Ltd.

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INTRODUCTION

The purpose of stormwater infiltration systems is to reverse or at least mitigate the negative effects of urbanization on hydrology by temporarily detaining runoff, increasing infiltration to groundwater and decreasing runoff to downstream receiving waters. Stormwater infiltration systems typically consist of a pit in the ground filled with gravel or sandy loam (see Figure 1). Most are long trenches or small rectangular soakaways (to fit into constrained urban environments), whereas larger systems may simply consist of a dry pond over freely draining soils (Fujita, 1994; Argue and Pezzaniti, 2003; Dechesne *et al.*, 2005).

Models of infiltration trenches require at least two-dimensional (2D) representation to accurately simulate infiltration through the bottom and the side walls, both of which may comprise a significant proportion of total outflows. Indeed, early design methods assumed the bottom will completely clog and all flows will occur through the side walls (BRE, 1991; Stahre and Urbonas, 1992; Mikkelsen *et al.*, 1996) as one-dimensional (1D) horizontal flow. Despite this, many existing models and guidelines including the MUSIC (v3) stormwater model (CASQA, 2003; CWP, 2003; Dussaillant *et al.*, 2004; Wong *et al.*, 2005) assume flows only occur through the

bottom as 1D vertical flow. A few guidelines and methods (Argue and Pezzaniti, 2003; Auckland City Council, 2003; Dechesne *et al.*, 2004) allow for quasi-2D conditions with vertical flow from the base and horizontal flow from the side walls. Most of these methods are based on sizing the storage to retain a design storm event and empty completely within a specified time. Most assume a constant infiltration rate (Wong *et al.*, 2005) or use soil equations such as the Horton or the Green–Ampt equation (Papa and Adams, 2005). Clogging is also a major concern (Lindsey *et al.*, 1992; Pokrajac and Deletic, 2002) and is sometimes approximated rather simplistically using a lower hydraulic conductivity value (Argue and Pezzaniti, 2003), such as 50% of the design value. Although better than ignoring clogging altogether, this method does not allow for the progressive build up of clogging over time and thus provides little insight into the timing of the infiltration system becoming ‘noncompliant’ with the adopted performance objectives.

Numerous 2D variable saturated models are available for simulating flow through soils. However, they are rarely applied in the design of stormwater infiltration systems because of their data and expertise requirements and complexity. Boundary conditions are also difficult as inclusion of the storage within the model may introduce numerical difficulties whereas exclusion produces a complex boundary condition. In one study, Duchene *et al.* (1994) compared a simple 1D model that uses Darcy’s law with a unit hydraulic gradient to calculate 1D saturated

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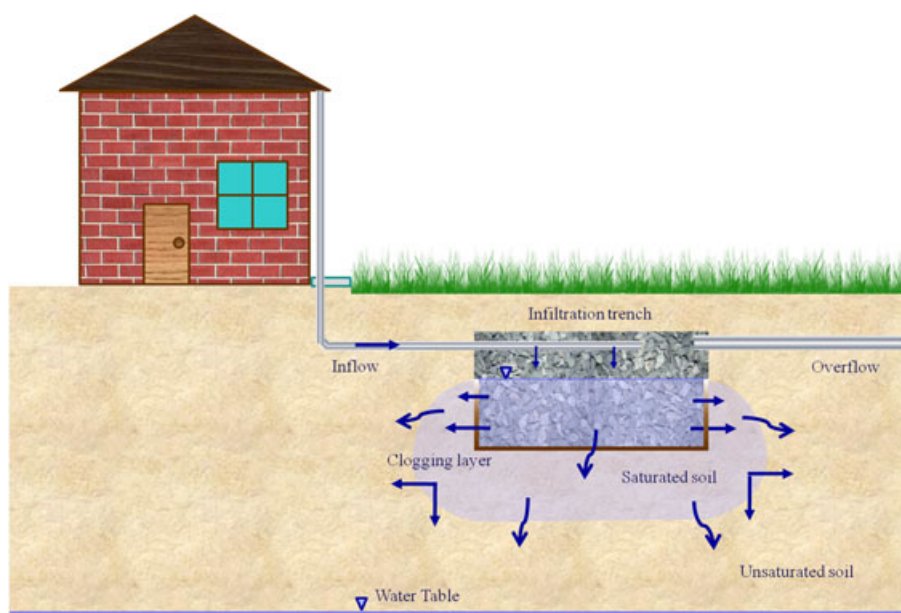


Figure 1. Typical infiltration trench

infiltration through the bottom and sides of the trench separately, with a 2D finite element model. Both were only applied for constant water level conditions in a stormwater infiltration trench. The simple model underestimated flows by 24% to 54%, with the error increasing with storage depth. The often-used assumption of 1D vertical saturated flow would be even more conservative. Duchene also compared 1D saturated flow through the side of the trench with a 2D simulation assuming an artificial clogging layer of low hydraulic conductivity, finding an assumption of no flow through the bottom and 1D horizontal flow would underestimate flows by at least 50% for the scenario considered. These results illustrate the potential inadequacies of simple models and approximate assumptions for systems such as infiltration trenches with 2D flow patterns. Clearly, there is a need for a 2D model of stormwater infiltration systems to simulate the hydraulic behaviour of the storage and soils that is effective and simple to use.

This article presents the development of such a model, the relevant aspects of 2D laboratory experiments, and the calibration and validation of the model using the laboratory data. The model presented is a further development of a 1D stormwater infiltration (Browne *et al.*, 2008). The model is unique in that it uses a soil moisture-based solution for the Richards equation yet can handle saturated flows with significant ponding depths. The result is an integrated model that can simulate flow through the storage and into the surrounding soil and can handle very dry soils with steep soil characteristic curves underlying the storage, which are common conditions for infiltration systems.

MODEL DEVELOPMENT

A 2D representation of a typical stormwater infiltration system is provided in Figure 2. If there is enough inflow, water ponds in the storage, water infiltrates into the soil

both vertically and horizontally and a saturated zone forms in the surrounding soils after prolonged ponding. The model must be able to model early wetting conditions with high pressures and rapid inflows into dry soils. If a clogging layer develops at the gravel–soil interface, the saturated zone will be constrained and eventually disappear, whereas unsaturated flow processes will dominate for conditions with low flows or clogging.

The previously mentioned model of the system presents challenges even for typical variably saturated flow models such as SEEP/W (Krahn, 2004) or Hydrus (Simunek *et al.*, 2006). As suggested by Krahn (2004), the gravel storage would preferably be excluded from the model to avoid numerical difficulties, and then a switching surface flux/head boundary could be used for the interface with the soil. However, switching boundaries implemented in existing models (e.g. SEEP/W Krahn, 2004) and SWAP (Kroes and Van Dam, 2003) are generally intended for 1D surface flow, whereas a more complex 2D switching flux/head boundary is required here. Alternatively, the gravel could be included as a coarse soil within the model, similar to the approach used by Gobel *et al.* (2004). However, this approach was not adopted as the combination of coarse soils and the clogging layer at the gravel–soil interface will likely cause significant numerical difficulties when the clogging model is implemented.

Conceptualization and governing equations

A 2D model is conceptualized with four major blocks representing the storage, the clogging layer, the saturated soil zone surrounding the storage and the unsaturated soil zone (Figure 2). This article focuses on the storage and soil zone modules, whereas the clogging module will be discussed in a subsequent publication. The key equations are presented in Figure 2 and discussed in the following sections.

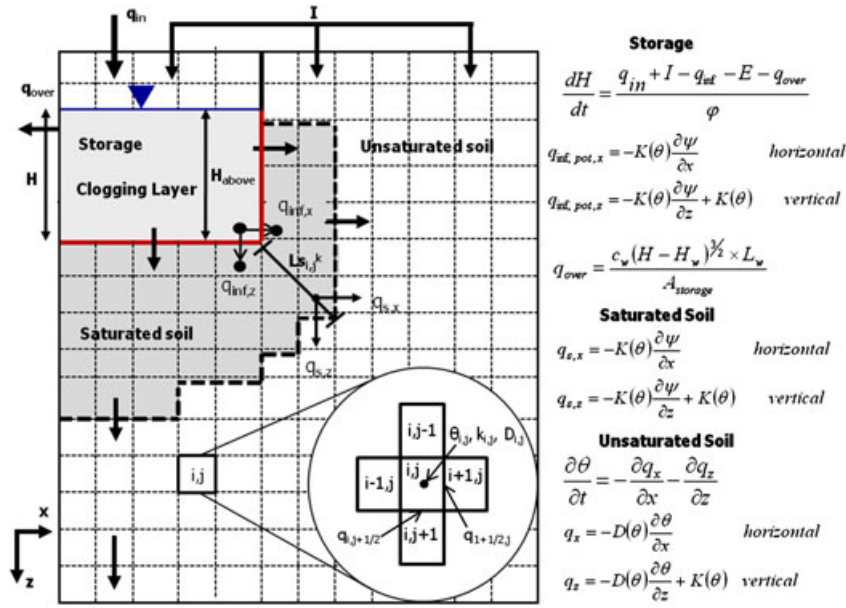


Figure 2. Conceptualization and equations for a stormwater infiltration system in 2D

Unsaturated soil zone. Flow through unsaturated soils is simulated using the Richards equation (Celia *et al.*, 1990) expressed in its soil moisture–based form:

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q_x}{\partial x} - \frac{\partial q_z}{\partial z} \quad (1)$$

Where,

$$\begin{aligned} q_x &= -D(\theta) \frac{\partial \theta}{\partial x} \\ q_z &= -D(\theta) \frac{\partial \theta}{\partial z} + K(\theta) \end{aligned} \quad (2)$$

where t is the time (s), x and z are the horizontal and vertical coordinates (Figure 2), respectively (m), q is the Darcy flux ($\text{m}^3/\text{s}/\text{m}^2$), θ is soil moisture (–), $K(\theta)$ is hydraulic conductivity (m/s) as a function of soil moisture, diffusivity is defined as $D(\theta) = K(\theta) \partial(\psi) / \partial \theta$ (m^2/s) and pressure head is defined as $\psi = p/\rho g$ (m), where p is pressure (N/m^2), ρ is water density (kg/m^3) and g is acceleration due to gravity m/s^2 .

The moisture-based form of the Richards equation was adopted because it was anticipated that freely draining granular soils with steep soil characteristic curves, which surround the system, will become very dry during long interevent periods common under Australian conditions. For modelling infiltration into dry soils with steep wetting fronts, conventional pressure head–based models may require fine spatial and temporal discretization to avoid convergence failures. This may result in long run times, and soil moisture–based front-tracking models may be more efficient and robust for this problem (Huang *et al.*, 1996; Grifoll and Cohen, 1999).

Saturated soil zone. A simple wetting front-tracking algorithm was combined with Darcy’s law to approximate flow through a saturated zone assumed to extend as a

continuous front from the storage (Figure 2). It is assumed that this zone is fully saturated and inflows and outflows are equal. Saturated fluxes at the saturated–unsaturated interface are calculated according to Darcy’s law for horizontal and vertical directions, respectively, as follows:

$$\begin{aligned} q_{s,x} &= -K_s \frac{\partial \psi}{\partial x} \quad \text{horizontal} \\ q_{s,z} &= -K_s(\theta) \frac{\partial \psi}{\partial z} + K(\theta) \quad \text{vertical} \end{aligned} \quad (3)$$

where $q_{s,x}$ and $q_{s,z}$ are the flow rates from the saturated zone to the unsaturated zone in the horizontal and vertical directions, respectively (m/s) (see Figure 2), and k_s is the saturated hydraulic conductivity (m/s).

Storage. The storage is simulated using a 1D reservoir equation:

$$\frac{dH}{dt} = \frac{q_{in} + I - q_{inf} - E - q_{over}}{\varphi} \quad (4)$$

where H is the depth of water in the storage from the storage base (m), and φ is the porosity of the storage media (–). Flow rates ($\text{m}^3/\text{s}/\text{m}^2$) are defined as follows: q_{in} is the stormwater inflow rate, I is the rainfall intensity, q_{inf} is the soil infiltration rate (potential infiltration volume through the bottom and sides divided by the storage plan area), E is the evaporation rate from the storage and q_{over} is the weir overflow rate from the storage. The soil infiltration, q_{inf} , is limited to the minimum of the potential infiltration rate, $q_{inf,pot}$, and the available volume of water (per time step), $q_{available}$, as follows:

$$q_{inf} = \min(q_{available}, q_{inf,pot}) \quad (5)$$

The potential infiltration rate, $q_{\text{inf, pot}} \text{ (m}^3/\text{s/m}^2\text{)}$, for saturated or unsaturated flow from the storage to the soil is calculated according to Equation (3), substituting $q_{\text{inf, pot, x}}$ and $q_{\text{inf, pot, z}}$ for q .

It is assumed that the system has an overflow weir or outlet structure to allow for flows in excess of the infiltration and storage capacity. The weir overflow rate, $q_{\text{over}} \text{ (m}^3/\text{s/m}^2\text{)}$, is calculated as follows:

$$q_{\text{over}} = \frac{C_w(H - H_w)^{3/2} \times L_w}{A_{\text{storage}}} \quad (6)$$

where C_w is the weir coefficient, H_w is the level of the weir above the bottom of the storage (m), L_w is the length of the weir (m) and A_{storage} is the plan area of the storage (m^2).

The integration of the storage within the soil flow model was considered. However, simulating flow through coarse gravel, a clogging layer then dry surrounding soils with steep soil characteristic curves creates a challenging problem. Initial efforts suggested it would be time consuming and problematic to solve numerically, and therefore a storage reservoir model was integrated within the model instead. This allows simultaneous consideration of the storage and soil to determine infiltration into the soil in two dimensions, that is, a time-variable 2D flux/head boundary. The user can simply supply an inflow flux while the model calculates flows from the storage model to the soil flow model. This approach provides support for the later integration of modules for filtration of sediment, the development of a clogging layer and reductions in flow rates.

Parameters and boundary conditions

The soil characteristic curves relating soil moisture, pressure head, hydraulic conductivity and diffusivity are predicted using the van Genuchten–Mualem soil model (Mualem, 1976; van Genuchten, 1980). The inputs are as follows: θ_s is the saturated moisture content, θ_r is the residual moisture content, α and n are the curve fitting parameters, l is the pore connectivity parameter and K_s is the saturated hydraulic conductivity. The Brooks–Corey, Vogel and Cislérova soil models or a table of externally generated soil hydraulic properties can alternatively be used, providing considerable flexibility in input soil parameters.

The time series inputs to the model are rainfall intensity over the model surface, I , stormwater inflow rate, q_{in} , and storage evaporation, $E \text{ (m}^3/\text{s/m}^2\text{)}$. A time-variable storage depth (m), where the model calculates the required stormwater inflow rate to maintain the specified depth, can be input instead of the stormwater inflow rate.

At the interface between the storage and the soil, a switching flux/head boundary condition is used. When inflows initially occur and while they are less than the outflow, a flux boundary is applied to the soil. If at some point the inflow exceeds outflow, then the model will calculate the depth of ponding that occurs in the storage. Once a depth of ponding greater than zero is calculated,

the model applies a head boundary condition to the soil with flows calculated for all nodes up to the water level. This is maintained until the water level falls to zero, at which time a flux boundary is again applied. The method is similar to the 1D switching boundary in SWAP (Kroes and Van Dam, 2003) and is applied in two dimensions to both the bottom and the side of the storage in this study, extending the range of the model beyond existing practice.

It is assumed that generally half of an infiltration trench will be modelled, taking advantage of the symmetry of typical systems. Therefore, the left boundary (Figure 2) is a no-flow boundary. The right boundary can be specified to be either a no-flow boundary or a transmissive boundary, assuming that water can freely flow out of the model along the boundary. The lower boundary can be either freely draining or restricted to the saturated hydraulic conductivity of the underlying soil (subsoil).

Numerical solution

The modelled region is discretized using a simple Godunov-type finite volume scheme (Hirsch, 2007), in which the region is represented using a grid of cells with averaged soil characteristics for each cell and an approximation is used to estimate fluxes across cell interfaces. The cells are numbered $i = 1$ to P in the horizontal direction (left to right) and $j = 1$ to J in the vertical direction (top to bottom). A square grid of fixed size is used, although there is potential to vary the sizing if required. The characteristics of the soil including soil moistures and pressure heads are evaluated at nodes in the centre of each cell and are averaged across the cell. Hydraulic conductivities and diffusivities at the intervals between two nodes are evaluated as arithmetic means of the nodal values. The arithmetic mean hydraulic conductivity is generally reliable for problems of infiltration into dry soils and used in popular variably saturated flow models, whereas geometric and harmonic means are typically less accurate (Parissopoulos and Wheeler, 1988; Van Dam and Feddes, 2000; Simunek *et al.*, 2006). This was confirmed through limited testing of alternatives for cases of infiltration into dry soils, which showed the arithmetic mean to more reliably produce an acceptable approximation. An explicit noniterative solution with variable time steps is used. The numerical solution was adapted from Tuteja *et al.* (2004), with all calculations undertaken for each column of cells from left to right and each cell within a column from top to bottom sequentially, calculating flows first then soil moistures.

Fluxes across cell boundaries are calculated by discretizing Equation (1) as

$$\theta_{i,j}^{k+1} = \theta_{i,j}^k - \left[\frac{q_{i+1/2,j}^k - q_{i-1/2,j}^k}{\Delta x} \right] \Delta t - \left[\frac{q_{i,j+1/2}^k - q_{i,j-1/2}^k}{\Delta z} \right] \Delta t \quad (7)$$

where,

$$\begin{aligned} q_{i+1/2,j}^k &= - \left[\frac{D(\theta_{i+1,j}^k) + D(\theta_{i,j}^k)}{2} \right] \times \left(\frac{\theta_{i+1,j}^k - \theta_{i,j}^k}{\Delta x} \right) && \text{horizontal} \\ q_{i,j+1/2}^k &= - \left\{ \frac{D(\theta_{i,j+1}^k) + D(\theta_{i,j}^k)}{2} \right\} \times \left[\frac{\theta_{i,j+1}^k - \theta_{i,j}^k}{\Delta z} \right] + \left\{ \frac{K(\theta_{i,j+1}^k) + K(\theta_{i,j}^k)}{2} \right\} && \text{vertical} \end{aligned} \quad (8)$$

where $\theta_{i,j}^k$ is the moisture content (–) for node ij at time k ; $K(\theta_{i,j}^k)$ is the hydraulic conductivity (m/s) for node ij at time k ; $D(\theta_{i,j}^k)$ is the diffusivity (m²/s) for node ij at time k ; Δt is the time step (s); and Δx and Δz are the block thickness in the horizontal and vertical directions, respectively (m) (Figure 2).

Unsaturated and saturated zones. A front-tracking algorithm determines the extent of the saturated zone (which extends as a continuous wetting front from the storage) at each time step. A node is saturated if the soil moisture exceeds the saturation threshold soil moisture (typically 95%–99.99% saturation) and either

- the node immediately above or left of the node is saturated, or
- the node is immediately below or beside the storage and $H_{\text{above}} > 0$, where H_{above} is the depth of water in the storage above the centroid of the node (see Figure 2).

When saturated flows through the interface exceed unsaturated flows away from the interface, the saturated zone gradually extends downwards and outwards as subsequent layers are progressively saturated. Nodes within the saturated zone are assumed to be completely saturated.

When the depth of water in the storage falls, desaturation begins to occur and the saturated zone retreats. A node becomes unsaturated if any of the following occur:

- The depth of water in the storage falls below the elevation of the node centroid.
- The moisture content in an unsaturated node below or beside a saturated node decreases, indicating that unsaturated flows away from the interface exceed saturated flows through the interface.
- The depth of water becomes zero in which case all nodes become unsaturated.

Once the wetting front is tracked, the distance from the nearest point of the storage to the saturated–unsaturated zone interface, $L_{s,i,j}$ (Figure 2), is calculated for each saturated node at the interface. For nodes directly below or adjacent to the storage, this is calculated as the vertical or horizontal distance to the furthest cell edge, whereas

for nodes diagonally below the storage, it is calculated to

the furthest cell corner. A pressure head is then linearly interpolated for all saturated nodes adjacent to the saturated–unsaturated zone interface, assuming a linear decline in pressure from the storage to interface:

$$\psi_{i,j}^k = \frac{H_{\text{above}}^{k+1}}{L_{s,i,j}^k} \left(L_{s,i,j}^k - \sqrt{\Delta x^2 + \Delta z^2} \right) \quad (9)$$

where $\psi_{i,j}^k$ is the pressure head at saturated node, Δx is the horizontal distance from the storage to the node in the centre of the last saturated cell, Δz is the vertical distance from the storage to the node in the centre of the last saturated cell, k is the previous time step and $L_{s,i,j}^k$ is the distance between the storage–soil interface and the saturated–unsaturated surface interface corresponding to node ij at time k .

The flows through the saturated–unsaturated zone interface are calculated using the pressure heads for these adjacent saturated nodes according to Equation (3), discretized as

$$\begin{aligned} q_{s,x,i+1/2,j}^{k+1} &= -K_{s,i,j} \frac{0 - \psi_{i,j}^k}{\Delta x/2} && (10) \\ &= K_{s,i,j} \frac{\psi_{i,j}^k}{\Delta x/2} && \text{horizontal} \\ q_{s,z,i,j+1/2}^{k+1} &= -K_{s,i,j} \left[\frac{0 - \psi_{i,j}^k}{\Delta z/2} \right] + K_{s,i,j} \\ &= K_{s,i,j} \left[\frac{\psi_{i,j}^k}{\Delta z/2} + 1 \right] && \text{vertical} \end{aligned}$$

where $q_{s,x,i+1/2,j}^{k+1}$ and $q_{s,z,i,j+1/2}^{k+1}$ are the horizontal and the vertical flow rates through the saturated–unsaturated zone interface, respectively (m/s), and K_s is the saturated hydraulic conductivity (m/s). This is an approximation, effectively assuming homogeneous soils throughout the saturated zone and a linear change in pressure head through the saturated–unsaturated zone interface between the nearest nodes. Given the limited information typically available during the design of stormwater infiltration systems, these assumptions should be sufficient for most cases. The effect of the capillary pressure at the wetting front is neglected in this equation as it is adequately represented through the unsaturated flow calculations as discussed by Browne *et al.* (2008).

Storage. The model solves Equation (4) numerically to obtain the depth of water in the storage at time step $k+1$, d^{k+1} , using the inflows and outflows from the same time step, $k+1$, and depth from the previous time step, k .

The soil infiltration, q_{inf} , depends on the potential infiltration rate and the available water volume, calculated as follows:

- a) Available infiltration rate, $q_{\text{available}}^{k+1}$ (m/s), the maximum infiltration rate if all of the available volume of water infiltrates within the time step,

$$q_{\text{available}}^{k+1} = \left[\frac{(H^k \times \varphi)}{\Delta t^{k+1}} + q_{\text{in}}^{k+1} + I^{k+1} - E - q_{\text{over}} \right] \quad (11)$$

- b) The method for potential infiltration depends on whether a saturated zone had formed at the previous time step with the discretizations of Equation (3) for unsaturated and saturated flow shown in Equations (12) and (13), respectively,

$$q_{\text{inf,pot},i,j}^{k+1} = \begin{cases} -K(\theta_{i,j}^k) \left[\frac{\psi(\theta_{i,j}^k) - H_{\text{above},i,j}^k}{\Delta x_i/2} \right] & \text{for node } i,j \text{ beside storage} \\ -K(\theta_{i,j}^k) \left[\frac{\psi(\theta_{i,j}^k) - H^k}{\Delta z_j/2} \right] + K(\theta_{i,j}^k) & \text{for node } i,j \text{ beneath storage} \end{cases} \quad (12)$$

$$q_{\text{inf,pot},i,j}^{k+1} = \begin{cases} K_{s,i,j} \frac{H_{\text{above},i,j}^k}{Ls_{i,j}^k} & \text{for node } i,j \text{ beside storage} \\ K_{s,i,j} \left[\frac{H_{i,j}^k}{Ls_{i,j}^k} + 1 \right] & \text{for node } i,j \text{ beneath storage} \end{cases} \quad (13)$$

where $Ls_{i,j}^k$ is the distance between the storage–soil interface and the saturated–unsaturated surface interface corresponding to node i,j (taken as the interface in the horizontal or vertical direction for nodes beside or underlying the storage, respectively).

Where a saturated zone exists, the storage outflow volume was adjusted to be consistent with the net outflow volume from the saturated zone to ensure mass balance as follows:

$$q_{\text{inf,act},i,j}^{k+1} = q_{\text{inf,pot},i,j}^{k+1} \times \frac{V_{\text{out,satzone}}^k}{V_{\text{out,storage}}^{k+1}} \quad (14)$$

where, $q_{\text{inf,act},i,j}^{k+1}$ (m/s) is the adjusted outflow rate from the storage to the saturated zone, $V_{\text{out,satzone}}^k$ is the total outflow volume from the saturated zone at time step k and $V_{\text{out,storage}}^{k+1}$ is the potential outflow volume from the storage to the soil at time $k+1$.

Boundary conditions. The input data consist of rainfall, I , stormwater inflow, q_{in} , and evaporation, E (all in $\text{m}^3/\text{s}/\text{m}^2$), at any variable time step, allowing data from a continuous rainfall–runoff or stormwater system model to be directly used. The model time step lengths are independent of inputs. Where a model time step spans more than one user input time step, I^{k+1} , E^{k+1} and q_{in}^{k+1} are calculated using a time-weighted average. Alternatively, a time-variable depth can be specified for comparison with fixed depth laboratory experiments with q_{in} calculated such that d^{k+1} is equal to the user specified depth at the corresponding time.

Outflows from the lower boundary are calculated as

$$q_{i,1+1/2}^{k+1} = \begin{cases} K(\theta_{i,j}^k) & \text{unsaturated} \\ K_{s,i,j}^k \times \left(\frac{H^{k+1}}{Ls_{i,j}^k} + 1 \right) & \text{saturated} \end{cases} \quad (15)$$

The outflow from the lower boundary is restricted to the hydraulic conductivity of the underlying subsoil,

K_{subsoil} (m/s). If a seepage face boundary is defined, outflows are zero when the boundary node is unsaturated. Outflows from the right side boundary are set to zero where it is a no-flow boundary, and outflows from each layer are set equal to the inflow to the last node in the layer for a transmissive boundary:

$$q_{P,j}^{k+1} = \begin{cases} 0 & \text{no flow boundary} \\ q_{P-1,j}^{k+1} & \text{transmissive boundary} \end{cases} \quad (16)$$

where P is in x space and is the last (right-hand side) node in the model domain.

Time step. An adaptive time stepping method first proposed by Staple (1966) accommodates the range of conditions from dry periods to sudden storms with rapid flows and saturation over dry soils. The resulting formulations for maximum

allowable time step length for time k , $\Delta t_{\max}^k$, and the predicted largest allowable time step for time $k+1$, $\Delta t_{\max}^{k+1}$, used here are as follows:

$$\Delta t_{\max}^k = \min \left\{ \left[r \frac{(\Delta z)^2}{D(\theta_{i,j}^k)} \right]_{j=1}^{j=J} \right\}_{i=1}^{i=P} \quad (17)$$

$$\Delta t_{\max}^{k+1} = \Delta t_{\max}^k + F \cdot [\Delta t^k - \Delta t^{k-1}] \quad (18)$$

where $\Delta t_{\max}^k$ is the maximum allowable time step length for time k and $\Delta t_{\max}^{k+1}$ is the time step for time $k+1$. $D(\theta_{i,j}^k)$ is the diffusivity for node i,j for time k , r is the Staple criterion for stability and F is a multiplier to accelerate the change in time step (typically around 1.2). $\Delta t_{\max}^{k+1}$ is also minimized at the onset of ponding. Values of r from 0.15 to 0.3 used by Staple (1966) were found to be satisfactory. As the nonlinearity of the soil curve increases, r can be decreased or F increased to ensure that the time step size remains within the range required for stability.

EXISTING NUMERICAL BENCHMARK AND EXPERIMENTAL DATA

The model results were compared with the results of another model and measurements from a control laboratory study, both of which are discussed in the following sections.

Numerical benchmarking

The model was compared with the SWMS model (Simunek *et al.*, 1994) used in Hydrus. The example considers a nominal storage with constant depth and 2D flow into the surrounding soil. The model domain has a depth of 1.3 m and a width of 1.25 m. A fixed grid size of 0.025 m was used for the stormwater infiltration model, and a variable grid size of 0.02 to 0.1 m was adopted for SWMS. A constant depth of 7.5 mm was applied to the storage. Soil parameters for the Vogel and Cislrova model are shown in Table I.

Laboratory data

A 2D test rig (Figure 3) was constructed at Monash University to study the behaviour of flows through and around a typical infiltration trench under controlled

conditions. The rig consisted of a steel frame with a wooden backing and removable clear Perspex segments on the front for visual inspection and repacking of the soil. It was 2.15 m wide $\times$ 2.1 m high $\times$ 0.25 m deep. The rig was filled with soil, and an infiltration trench (storage) was constructed as a gravel-filled section (0.5 m width $\times$ 0.8 m height) in the top left corner of the rig. A high hydraulic conductivity geotextile between the soil and the gravel prevented soil migration into the gravel. The gravel has a porosity of 0.48 and a median particle diameter of 10.5 mm. Twelve pressure sensors were installed in and around the trench. Time domain reflectometry (TDR) probes were installed to monitor moisture content in the soil and are discussed below. Inflows to the storage were monitored by recording the water level in the supply tank (Figure 3). Outflows were monitored using a rain gauge. For further details, see Siriwardene *et al.* (2007). Three experiments were used as follows:

Experiment A. The rig was packed with Unimin 50-N sand, a fine sand with a d_{15} of 130 μm , a d_{50} of 220 μm and a d_{85} of 290 μm . The sand was manually compacted with a heavy metal plate. Six TDR probes were used. To simulate multiple storm events, the storage was filled with clean water to a depth of 75 cm then allowed to drain to 5 cm. This process was repeated several times for 6 h each day on weekdays from 17/1/2005 to 26/1/2005. The data from 25/1/2005 were chosen for calibration and from 21/1/2005 for validation as these were representative of typical conditions and had clean data sets without any anomalies. Data for depth from a pressure probe in the storage and soil moisture at probe 12 (T12 in Figure 3) were used. The inflow rate for this experiment was estimated based on pressure data recorded for the storage during filling as the water level in the supply tank was not monitored. The volume of water needed to refill the storage was calculated from the gravel porosity and the recorded change in depth.

Experiment B. The rig was packed with a mixed loamy sand soil consisting of 90% 50-N sand and 10% sandy loam and highly compacted using a mechanical vibration compactor. The TDR probes were the same as experiment A. The storage was filled to a depth of 74 cm then allowed to drain to 7 cm repeatedly for approximately 7 h each weekday from 12/10/2005 to 21/10/2005 and from 26/10/2005 to 28/10/2005. Constant water level was maintained from 21/10/2005 to 25/10/2005. Data for 14/10/2005 were chosen for calibration and from 26 to 28/10/2005 for validation. Data for depth in the storage and soil moisture at probe 3 (T3 in Figure 3) were used for calibration and validation.

Experiment C. The rig was packed as for experiment B. Moisture content was monitored using 8 TDR probes. The storage was filled to a depth of 74 cm then allowed to drain to 7–10 cm. The cycles were repeated for approximately 7 h each weekday from 28/3/2006 to 1/4/2006 and again from 24/4/2006 to 28/4/2006. The periods 29/3/2006 and 1/4/2006 for were chosen for calibration and validation, respectively. Data for depth in the storage and

Table I. Vogel and Cislrova soil parameters for benchmark comparison

Parameter	Value
θ_r, θ_a	0.02
θ_s, θ_m	0.35
α (1/m)	4.1
n	1.964
K_s (m/s)	7.22×10^{-6}
θ_k	0.2875
K_k (m/s)	6.95×10^{-6}

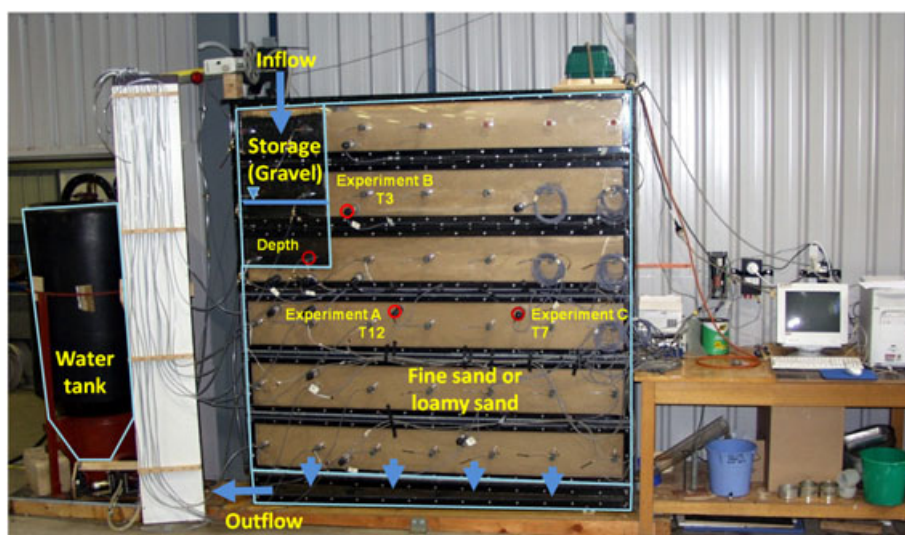


Figure 3. Two-dimensional experimental rig at Monash University (Browne, submitted 2010)

soil moisture at probe 7 (T7, Figure 3) were used for calibration and validation.

Soil tests were conducted to obtain a soil characteristic curve for the mix of sand and sandy loam used in experiments B and C. A Tempe cell (Fredlund and Rahardjo, 1993) was set up with a compacted sample of disturbed soil taken from the rig following experiment C. A multiple-step pressure test (van Dam *et al.*, 1994) for a range of soil moistures was undertaken. Suctions were applied to the outlet at small increments to discretize the curve for the coarse soil. First, the sample was saturated from the bottom-up, and a 1.5-m flexible outlet pipe filled with water was connected. The level of the outlet pipe relative to the Tempe cell was lowered incrementally to provide the fine discretization needed. Pressure was then also applied following the largest suction. Suctions and pressures were applied until equilibrium was reached with no further outflow occurring for each increment. A beaker containing outflows was then weighed to obtain moisture content. The final moisture content was determined by oven drying the sample. A one-step pressure test (van Dam *et al.*, 1992) at 200 kPa on a fresh sample was used to obtain an estimated residual moisture content of 4.7%.

The Retention Curve (RET) program (van Genuchten *et al.*, 1991) was used to fit van Genuchten soil curves to the experimental soil retention data using a four-parameter fit to θ_r , θ_s , α and n and a three-parameter fit using $\theta_r = 0.047$. The resulting soil curves are shown in Figure 4. The curve is very steep as expected for sand, whereas the air entry value is conversely relatively high because of the loam content. The four-parameter fit resulted in a residual moisture content of 0. The three parameter fit with measured $\theta_r = 0.047$ was adopted for modelling to evaluate the model's performance using soil parameters obtained from the independent laboratory tests and used as a starting point for calibration.

For the 50-N sand used in experiment A and the loamy sand mix used in experiments B and C, the saturated hydraulic conductivity, K_s , was determined on disturbed

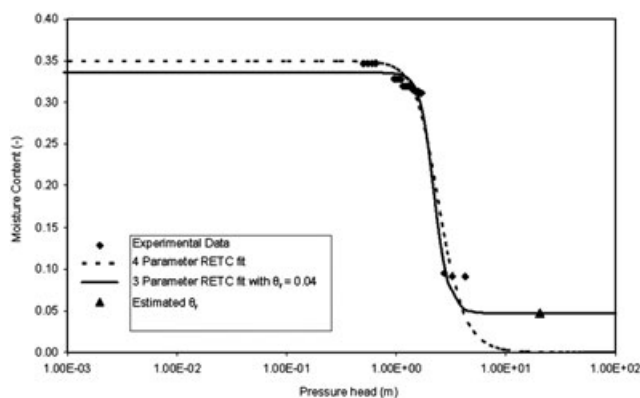


Figure 4. Soil characteristic curves for 90% 50-N sand, 10% sandy loam soil used in experiments B and C

samples using the standard laboratory test (AS 4419–2000). It was found that K_s was 3.43×10^{-4} m/s and 7.81×10^{-5} m/s for the sand and the loamy sand mix, respectively. As disturbed samples were used, these K_s values were adopted only as approximate initial estimates.

MODEL TESTING

Testing the model against SWMS results was relatively straightforward. The models were run to simulate a period of 15 h. The location of the wetting front based on soil moistures was plotted at intervals to compare the models.

Testing the model against the laboratory data set required first calibration then verification. The calibration of the model parameters (θ_s , θ_r , α , n and K_s) was undertaken using the popular optimization tool PEST (Doherty, 2004). A grid size of 0.05 m and an adaptive time stepping with a range of up to 10 s were used. The 'r' parameter was set at 0.2 and 'F' at 1.2. Calibration for each of the three experiments was undertaken using five different combinations of observed data, as summarized in Table II and detailed below. Data for depth in the storage and for soil moisture at a selected monitoring

Table II. Calibration cases

Case	Model calibrated parameters	Calibration data
1	None	–
2	θ_r , θ_s , α , n and K_s	Depth
3	α and n	Depth
4	θ_r , θ_s , α , n and K	Depth and soil moisture
5	α and n	Depth and soil moisture

probe or cell in the soil (see Section 3.2) were used for calibration. Data at either a 2-min time step (experiment A) or a 1-min time step (experiments B and C) were used. The data for soil moisture spanned the period of calibration, and the data for depth were restricted to the period where depths exceeded zero. For a 1-day calibration, this resulted in 720/1440 soil moisture observations and approximately 200/400 depth observations.

1. Calibration using independently measured soil parameters. For either existing or planned systems, the soil characteristics of field samples can be obtained from laboratory tests as described earlier. Therefore, the model was initially run for all experiments using the soil parameters obtained in the laboratory for the three-parameter RETC fit with $\theta_r = 0.047$ (see Figure 4). An initial saturated hydraulic conductivity, K_s , of 1.33×10^{-5} m/s was used for experiments B and C and increased to 2.66×10^{-5} m/s for experiment A. The soil in the rig was highly compacted, and the initial saturated hydraulic conductivities were selected following preliminary simulations that indicated the conductivities in the rig were much lower than the standard laboratory measurements obtained for disturbed samples. These parameters were used as the initial parameters for all of the subsequent calibrations.
2. Calibration using storage depth data. For existing stormwater infiltration systems or pilot test pits, the depth of water in the storage media can readily be measured using a depth sensor in a borehole. Because the outflows commonly used for the calibration of laboratory column experiments are impractical in the field, depth is proposed as a substitute. The parameters obtained from laboratory soil testing may not be representative of field conditions because of disturbance, small sample size, variations in compaction and experimental complications such as air entrapment and macropore flow (Kool *et al.*, 1987; Hopmans *et al.*, 2002; Mertens *et al.*, 2005). Therefore, it may be advantageous if 'field' depth data can be used to directly calibrate soil parameters. The model was calibrated for five parameters, θ_r , θ_s , α , n and K_s , using depth for all three experiments.
3. Using storage depth data and independently measured soil parameters. Using a single data set (depth) may not provide sufficient information for the calibration of five parameters (Kool *et al.*, 1987). Therefore, the model was calibrated using PEST for just two parameters α and n using the measured depth, whereas the other parameters were taken from the RETC calibration to the laboratory data.

4. Using storage depth and soil moisture content data. Calibration studies have shown that it is desirable to have more than one type of data for the reliable calibration of soil parameters (Kool *et al.*, 1987; van Dam *et al.*, 1992; Durner *et al.*, 1999; Hopmans *et al.*, 2002). Therefore, calibration was undertaken for the five soil parameters (θ_r , θ_s , α , n and K) using both depth and moisture content data to provide a comparison with the above approaches.

5. Using storage depth, soil moisture and independently measured soil parameters. The model was calibrated for α and n using depth and moisture content with the independent estimate for the remaining parameters. This is the most data-demanding exercise and although less practical may result in the best model performance. For each model run, the results for modelled and measured depth, moisture content and outflow rate were plotted for visual comparison. The Nash–Sutcliffe efficiencies were calculated and used to assess the model reliability. The parameter correlation matrix from PEST was also examined to evaluate the identifiability of parameters and thus uniqueness of the solution (highly correlated parameters indicate that the solution may not be unique and many different parameter sets may produce a similar result for the objective function). The identifiability was considered good if parameter correlations were all less than 0.5, moderate if at least one was 0.5–0.6 and poor if at least one exceeded 0.6.

RESULTS AND DISCUSSION

Comparison with SWMS

The results from the comparison with SWMS are shown in Figure 5. It can be seen that the models produce fairly similar results for the position of the wetting front at the time intervals of 0, 5, 10 and 15 h. This provides confidence that the approach adopted is reasonable.

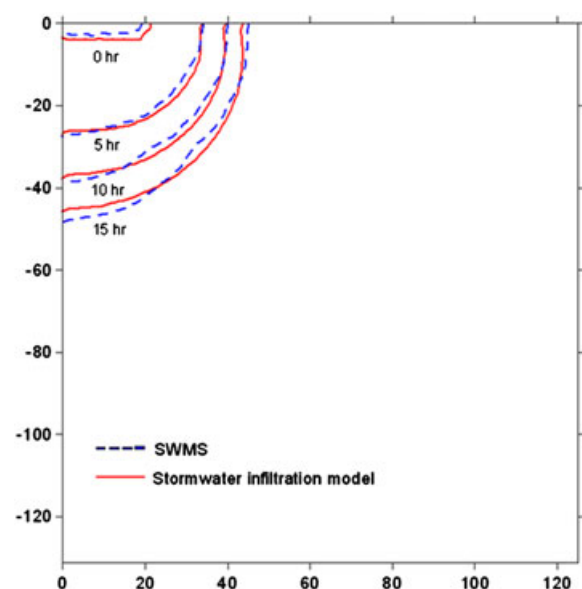


Figure 5. Comparison of model with SWMS for 2D infiltration

Comparison against laboratory data—calibration step

The soil parameters obtained from model calibration using PEST and also the direct fit to laboratory data using RETC are summarized in Table III. The resulting Nash–Sutcliffe Efficiencies are summarized in Table IV and calibration plots for experiment C are shown in Figure 6. A reliable model will have an efficiency approaching 1, whereas a result less than 0 indicates the mean of the observed values would provide a better prediction than the model (Verstraeten and Poesen, 2001).

1. Calibration results using independently measured soil parameters. The results for experiment C suggest the model follows the general trends but overestimates the peak depths and moisture contents and drains too slowly. For experiment B, the opposite is observed with the peaks being underestimated and the model draining too quickly. Generally, for these two experiments, the fit to depth is good with high Nash–Sutcliffe efficiencies whereas the fit to soil moisture is poor with low efficiencies. The fit to experiment A is poor as expected because the independent parameter set is for the mixed sandy loam soil, not the 100% 50-N sand used for this experiment.
2. Calibration results using storage depth data. As expected, the results show an improved fit to depth for all experiments. The fit to moisture content is poorer for experiment B but improved for experiments A and C, becoming moderate for experiment A. The identifiability of parameters is moderate for experiment A but poor for experiments B and C with at least three pairs of highly correlated parameters. Depth data alone would seem insufficient to obtain a unique calibration to the model's five parameters.
3. Calibration results using storage depth and independently measured soil parameters. For experiment C, the fit to depth was slightly improved at the expense of moisture content (the model slightly underestimates the peaks and drains more slowly than observed). For experiment B, no improved fit was obtained, suggesting the two parameters were at or near a global or local optimum. The two parameters show moderate levels of correlation.
4. Calibration results using storage depth and soil moisture content data. The results indicate that fits to depth for all experiments are similar to those achieved for the independent parameters. Although they are not quite as good as those obtained for calibration to storage depth alone, the fits to soil moisture are all substantially improved with a moderate fit to moisture content for experiment B and good fits for experiments A and C. For experiment A, the fit to depth is marginally worse than the independent parameters and much poorer than the five-parameter fit to depth alone with modelled peak depths falling well short of the measured peak depths. Comparing the results for other soil moisture probes for which no calibration was undertaken, this calibration provides the best fit between modelled and observed soil moisture content data for the other probes. Overall, this calibration is clearly the best overall fit to the observed data, except depth for experiment A.
5. Calibration results using storage depth, soil moisture and independently measured soil parameters. For experiments B and C, the fits to depth are good and to soil moisture are poor. For experiment C, the fit to soil moisture is worse than the independent parameter set but better than the calibrations to depth. For experiment B, the results show little change from the independent soil parameters.

Table III. Soil parameters

Parameter	Independent soil parameters	Five-parameter calibration to depth	Independent and two-parameter calibration to depth	Five-parameter calibration to depth and soil moisture	Independent and two-parameter calibration to depth and soil moisture
Experiment A					
θ_r	0.047	0.070	–	0.001	–
θ_s	0.335	0.3 ^a	–	0.3 ^a	–
α (1/m)	0.47	1.40	–	1.16	–
N	6.95	10 ^a	–	10 ^a	–
K_s (m/s)	5.21×10^{-5}	2.30×10^{-5}	–	3.15×10^{-5}	–
Experiment B					
θ_r	0.047	0.053	0.047	0.100 ^a	0.047
θ_s	0.335	0.400 ^a	0.335	0.307	0.335
α (1/m)	0.47	0.44	0.47	0.18	0.87
N	6.95	7.04	6.95	7.67	10.0 ^a
K_s (m/s)	1.33×10^{-5}	1.17×10^{-5}	1.33×10^{-5}	1.14×10^{-5}	1.33×10^{-5}
Experiment C					
θ_r	0.047	0.023	0.047	0.013	0.047
θ_s	0.335	0.4*	0.335	0.3*	0.335
α (1/m)	0.47	0.40	0.30	0.48	0.23
n	6.95	7.34	5.27	8 ^a	8 ^a
K_s (m/s)	1.33×10^{-5}	1.37×10^{-5}	1.33×10^{-5}	1.62×10^{-5}	1.33×10^{-5}

^a Parameter calibrated to calibration constraint.

Table IV. Nash–Sutcliffe Efficiency, RMSE and model identifiability for calibration runs

Parameter	Independent soil parameters		Five-parameter calibration to depth		Independent and two-parameter calibration to depth		Five-parameter calibration to depth and soil moisture		Independent and two-parameter calibration to depth and soil moisture	
	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE
Experiment A										
Depth	0.845	0.081	0.944	0.049	–		0.479	0.149	–	
Soil moisture (T12)	0.526	0.070	0.709	0.055	–		0.930	0.027	–	
Identifiability	Poor		Poor		–		Poor		–	
Experiment B										
Depth	0.979	0.028	0.981	0.026	0.983	0.025	0.880	0.066	0.901	0.060
Soil moisture (T4)	0.368	0.085	0.465	0.078	0.497	0.076	0.893	0.035	0.597	0.068
Identifiability	Poor		Poor		Good		Moderate		Good	
Experiment C										
Depth	0.976	0.030	0.984	0.025	0.982	0.027	0.963	0.026	0.887	0.066
Soil moisture (T7)	–0.358	0.072	0.324	0.051	–0.300	0.071	0.829	0.038	0.193	0.056
Identifiability	Poor		Poor		Good		Poor		Moderate	

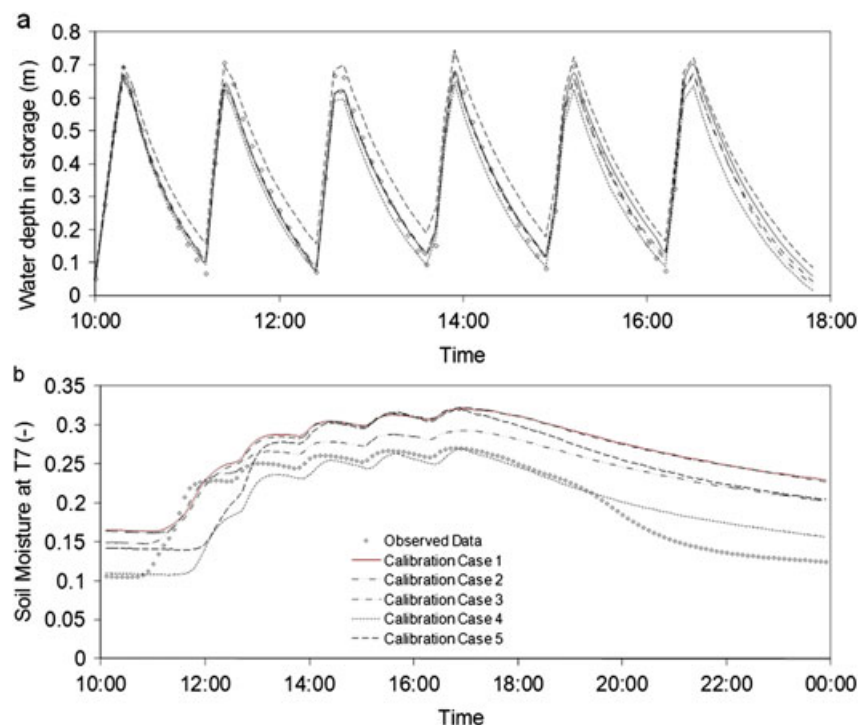


Figure 6. Calibration plots for experiment C with observed and modelled data. The first plot shows soil moistures at T7 (see Figure 3), and the second plot shows the depth of water within the storage over the period for which filling and draining occurred

Generally, depth is modelled reliably for most of the calibration approaches. Soil moisture is simulated with variable success with the five-parameter calibration using depth and soil moisture data providing the best fit. The results indicate that the representation of soil flows can be significantly improved through calibration to field data rather than depending entirely on laboratory data, although the fits to depth were still quite reasonable. Under calibration, θ_s tended to be maximized or minimized, indicating that it may be preferable to determine this parameter in the laboratory. This would also increase the uniqueness of the calibration.

Comparison against laboratory data—validation step

The validation results are summarized in Table V and for experiment C are plotted in Figure 7. Consistently, good fits to depth were obtained for all parameter sets and experiments. The efficiencies for the fits to soil moisture vary more widely than those for depth, with the best fits being for the five soil parameter calibration using depth and soil moisture data. As a further spatial validation check, the Nash–Sutcliffe efficiencies for experiment C for a range of soil moisture probes, for which only T7 was calibrated, are shown in Figure 8. The results indicate that the

Table V. Nash–Sutcliffe efficiency, RMSE and model identifiability for validation runs

Parameter	Independent soil parameters		Five-parameter calibration to depth		Independent and two-parameter calibration to depth		Five-parameter calibration to depth and soil moisture		Independent and two-parameter calibration to depth and soil moisture	
	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE	<i>E</i>	RMSE
Experiment A										
Depth	0.777	0.130	0.954	0.062	–	–	0.771	0.153	–	–
Soil moisture (T12)	0.337	0.070	0.485	0.062	–	–	0.780	0.029	–	–
Identifiability	Moderate		Poor		–		Moderate		–	
Experiment B										
Depth	0.948	0.045	0.961	0.039	0.966	0.036	0.782	0.092	0.948	0.045
Soil moisture (T4)	0.403	0.082	0.497	0.075	0.527	0.073	0.904	0.033	0.613	0.066
Identifiability	Poor		Poor		Good		Poor		Poor	
Experiment C										
Depth	0.972	0.030	0.965	0.028	0.963	0.030	0.930	0.059	0.944	0.046
Soil moisture (T7)	0.068	0.044	0.566	0.058	0.115	0.044	0.834	0.020	0.498	0.035
Identifiability	Poor		Poor		Poor		Poor		Poor	

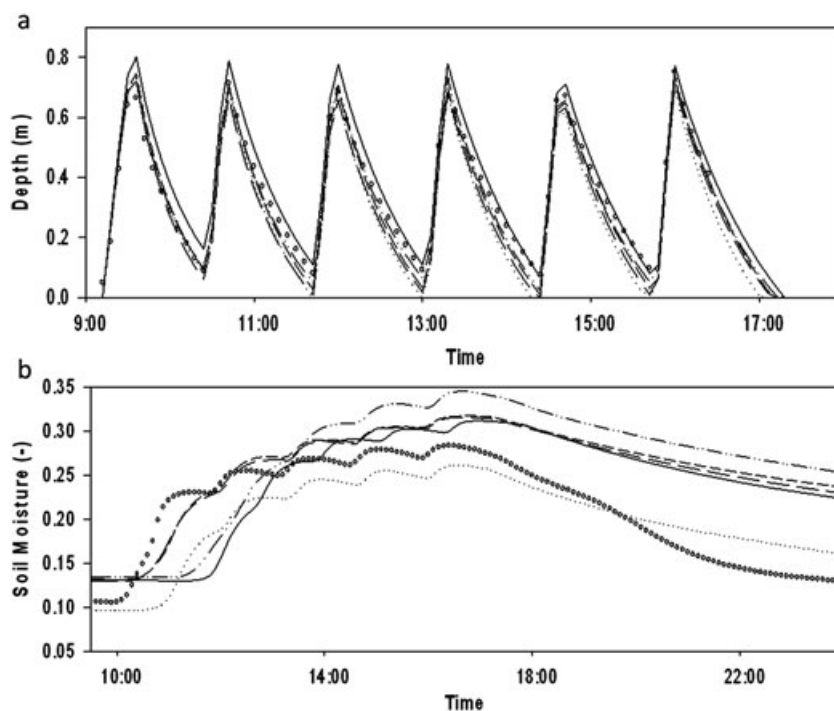


Figure 7. Validation plots for experiment C with observed and modelled data. The first plot shows soil moistures at T7 (see Figure 3), and the second shows depth of water within the storage over the period for which filling and draining occurred Figure 8 experiment C Nash–Sutcliffe Efficiencies for a range of moisture probes

model provided a good estimate of the spatial variability of soil moistures.

The calibration and validation exercise confirms that the model behaves as expected and is able to reasonably simulate typical conditions for a stormwater infiltration system, especially storage depths and draining times.

Practical implications

Laboratory-estimated soil parameters provided a good fit to depth but poor fit to soil moisture. The use of standard soil parameter sets such as those available in Carsel and Parrish (1988) would probably result in poorer

representation. Calibration to field measurements of soil moisture can improve the accuracy of the model and likely its capacity to simulate conditions that vary from those under which it was calibrated. However, these may be difficult to obtain, and it was found that field depth measurements and/or laboratory soil testing data will generally provide satisfactory estimates of depths and drainage times. Because these are of greatest interest for the simulation of stormwater infiltration systems, this may be sufficient in most cases. The benefit of calibrating to field data is that the parameters are calibrated to the *in situ* soils without disturbance and at an appropriate scale, avoiding issues that can arise with these during laboratory

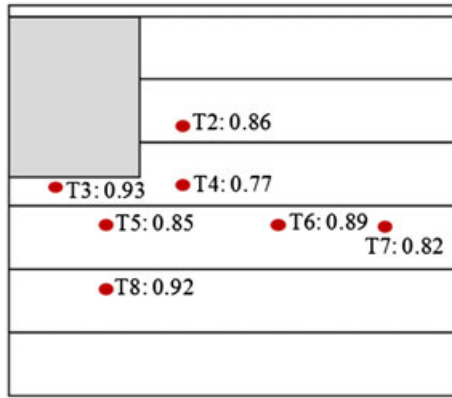


Figure 8. Results of experiment C with calibration to depth and soil moisture showing spatial distribution of Nash–Sutcliffe Efficiencies for a range of moisture probes. Note that only storage depth and soil moisture at T7 were calibrated

analysis. Therefore, the parameter set is likely to be more representative of field conditions. A single day of observation data was sufficient for calibration for the experiments examined. The drawback of field data is the need for equipment to monitor depths and soil moistures.

Calibration investigations also indicated that the spatial location of moisture probes for calibration is important. The probe should be located close to, but not beneath, the test pit or infiltration system in a freely draining location where a good range of moisture contents will occur. This range should extend from dry to saturated but avoid areas with positive pressures that may influence measurements.

CONCLUSIONS

In this study, a 2D model specifically designed for stormwater infiltration systems was developed, focusing on the simulation of rapid infiltration from a porous storage into initially dry and freely draining soils. The model combines a soil moisture–based Richards equation solution for unsaturated flow with Darcy’s law, a front-tracking method for saturated flow and a storage model for ponding and flow through a highly porous storage. The integration of the storage enables a 2D switching flux/head boundary condition to be simulated, which is beyond the range of boundary conditions handled by most existing models. It also simplifies user input requirements, allowing fluxes from a rainfall–runoff model or stormwater system model to be input directly. The soil moisture–based solution is inherently mass conservative and noniterative. The model is relatively simple mathematically yet robustly handles the highly nonlinear conditions of rapid infiltration into very dry coarse-grained soils without difficulty. The model allows the application of a soil moisture–based method to problems involving ponded saturated conditions that previously could only be solved with pressure-based methods. It may have potential applications for the simulation of irrigation furrows, ephemeral streams or where the simulation of 2D ponding is required and steep wetting fronts or coarse-grained soils occur. One-dimensional models commonly used for simulating stormwater

infiltration systems have been shown to be relatively inaccurate for the simulation of stormwater infiltration trenches (Duchene *et al.*, 1994). The model developed in this study allows a 2D approach that takes antecedent moisture conditions into account to be applied.

Soil parameters were obtained from laboratory soil tests and calibrated based on observed data from experiments conducted in a 2D rig. The calibration and validations runs showed that it is possible to obtain a set of soil parameters using field observations rather than laboratory soil tests. A single day of observations with a broad range of depths and soil moistures was sufficient to obtain a good calibration fit. This has important implications for practice as laboratory soil tests may not be representative of field conditions. Validation tests confirmed the model behaved as expected. Future application to the simulation of stormwater infiltration systems and comparison with other models as well as implementation of modules to account for filtration of sediment and development of a clogging layer are planned.

NOTATION

A_{storage}	plan area of the storage (m^2)
C_w	weir coefficient
$D(\theta)$	diffusivity (m^2/s)
g	acceleration due to gravity (m/s^2)
H	water depth in the storage (m)
$H_{\text{above},i,j}^k$	water depth above the centroid of the cell, i,j at time k
H_w	level of the weir above the bottom of the storage
E	evaporation rate from the storage ($\text{m}^3/\text{s}/\text{m}^2$)
F	a multiplier to accelerate the change in time step (–)
$\psi(\theta)$	pressure head (m)
I	rainfall intensity ($\text{m}^3/\text{s}/\text{m}^2$)
i and P	node and number of nodes in horizontal direction (left to right), respectively
j and J	node and number of nodes in vertical direction (bottom to top), respectively
$K(\theta)$	hydraulic conductivity (m/s)
K_s	saturated hydraulic conductivity (m/s)
K_{subsoil}	hydraulic conductivity of the underlying subsoil (m/s)
K	time step (–)
l	the pore connectivity parameter
$L_{si,j}$	distance from the storage–soil interface to the saturated–unsaturated surface interface for node i,j (m)
L_w	length of the weir
m	empirical constant in the van Genuchten equation
p	pressure (N/m^2)
ρ	water density (kg/m^3)
φ	porosity of the storage media (–)
q	flux ($\text{m}^3/\text{s}/\text{m}^2$)
$q_{\text{available}}$	available infiltration rate ($\text{m}^3/\text{s}/\text{m}^2$)

q_{in}	stormwater inflow rate ($m^3/s/m^2$)
q_{inf}	soil infiltration rate ($m^3/s/m^2$)
$q_{inf,pot}$	potential soil infiltration rate ($m^3/s/m^2$)
q_{over}	overflow rate from the storage ($m^3/s/m^2$)
$q_{soilcap}$	available soil moisture storage capacity ($m^3/s/m^2$)
$q_{s,x}$ and $q_{s,z}$	horizontal and vertical flow rates through the saturated–unsaturated zone interface respectively ($m^3/s/m^2$)
r	Staple's criterion for stability
t	time (s)
x, z	horizontal and vertical coordinates (m)
$V_{out,satZone}$	total outflow volume from saturated zone (m)
$V_{out,storage}$	potential outflow volume from storage to soil (m)
α and n	curve-fitting parameters
Δt	time step length (s)
θ	soil moisture (–)
θ_r	residual moisture content
θ_s	saturated moisture content (–)
Δx and Δz	block thicknesses in the horizontal and vertical directions, respectively (m)
Δt_{max}	maximum allowable time step length (s)

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